

- I. A. projections are inv. FALSE.
 B. homog. matrix in $\mathbb{E}^3$ is 4×4 . TRUE
 homog. coordinates $(x, y, z, 1)$.
 C. reflections are invertible TRUE.
 D. tensor prod. is always sym. FALSE.

II. $[A]_K = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$, $[B]_K = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$, $[C]_K = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$, $[D]_K = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$

✓ *). orthogonal proj. of point D on the plane containing ABC.

$\vec{AB} \times \vec{AC} = m$ = normal vector to the plane ABC.

$$\vec{AB} = B - A = (-1, 1, 1) - (-1, -1, 1) = (0, 2, 0)$$

$$\vec{AC} = C - A = (0, 2, -1) - (-1, -1, 1) = (1, 3, -2)$$

$$m = (0, 2, 0) \times (1, 3, -2) = \begin{vmatrix} i & j & k \\ 0 & 2 & 0 \\ 1 & 3 & -2 \end{vmatrix} = i \cdot \begin{vmatrix} 2 & 0 \\ 3 & -2 \end{vmatrix} - j \cdot \begin{vmatrix} 0 & 0 \\ 1 & -2 \end{vmatrix} + k \cdot \begin{vmatrix} 0 & 2 \\ 1 & 3 \end{vmatrix} =$$

$$m = i \cdot (-4) - 2k = (-4, 0, -2) \quad | : -2$$

$$\Rightarrow m = (2, 0, 1)$$

! plane through A with normal $m = (2, 0, 1)$, $A(-1, -1, 1)$

$$2(x+1) + 0 + 1 \cdot (z-1) = 0$$

$$\bar{u}: 2x + z + 1 = 0. \Rightarrow a_{u+1} = 1.$$

$$P_{\bar{u}}^{\perp}(D) = D - \frac{\vec{a} \cdot D + a_{u+1} \cdot a}{|\vec{a}|^2} \cdot \vec{a}$$

$\vec{a} = m = (2, 0, 1)$ normal vector of the plane $\bar{u}$.

$$a_{m+1} = 1$$

$$P_{\bar{u}}^{\perp}(D) = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} - \frac{2+1}{5} \cdot (2, 0, 1) = (0, 1, 2) - \frac{3}{5}(2, 0, 1) = \left(\frac{6}{5}, 1, \frac{4}{5}\right)$$

$$\vec{a} \cdot D = (2, 0, 1) \cdot (0, 1, 2) = 2 \cdot 0 + 0 \cdot 1 + 1 \cdot 2 = 2.$$

$$|\vec{a}|^2 = (\sqrt{2^2 + 0^2 + 1^2})^2 = (\sqrt{5})^2 = 5 = \vec{a} \cdot \vec{a} = (2, 0, 1) \cdot (2, 0, 1) = 5.$$

8) reflection of D in plane ABC parallel to $\vec{DA}$. $\rightarrow$ parallel refl.
line, dir. vector $\vec{DA}$, point D .

$$\pi: 2x + z + 1 = 0$$

$$a = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, a_{n+1} = 1, v = \vec{DA} = (-1, -2, -1) = D - A, D = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$\text{Ref}_{\pi, v}(D) = D - 2 \cdot \frac{a^T D + a_{n+1}}{a^T v} v$$

$$a^T D + a_{n+1} = (2, 0, 1) \cdot \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + 1 = 2 + 1 = 3.$$

$$a^T v = (2, 0, 1) \cdot \begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix} = -2 - 1 = -3.$$

$$\text{Ref}_{\pi, v}(D) = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} - 2 \cdot \frac{3}{-3} v = (0, 1, 2) + 2 \cdot (-1, -2, -1) = (-2, -3, 0).$$

10) matrix form of the orthogonal reflection in the plane ABC :

$$\text{Ref}_{\pi}^{\perp}(P) = \left(I - 2 \cdot \frac{a \otimes a}{|a|^2} \right) P - 2 \cdot \frac{a_{n+1}}{|a|^2} \cdot a.$$

$$|a|^2 = (\sqrt{2^2 + 0^2 + 1^2})^2 = (\sqrt{5})^2 = 5.$$

$$a_{n+1} = 1.$$

$$a \otimes a = m \otimes m = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

$$I - 2 \cdot \frac{1}{5} \begin{pmatrix} 4 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} \frac{8}{5} & 0 & \frac{4}{5} \\ 0 & 0 & 0 \\ \frac{4}{5} & 0 & \frac{2}{5} \end{pmatrix} = \begin{pmatrix} -\frac{3}{5} & 0 & -\frac{4}{5} \\ 0 & 1 & 0 \\ -\frac{4}{5} & 0 & \frac{3}{5} \end{pmatrix}$$

$$\text{Ref} = \begin{pmatrix} -\frac{3}{5} & 0 & -\frac{4}{5} \\ 0 & 1 & 0 \\ -\frac{4}{5} & 0 & \frac{3}{5} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} - 2 \cdot \frac{1}{5} \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} =$$

$$= \begin{pmatrix} -\frac{8}{5} \\ 1 \\ \frac{6}{5} \end{pmatrix} - \begin{pmatrix} \frac{4}{5} \\ 0 \\ \frac{2}{5} \end{pmatrix} = \begin{pmatrix} -\frac{12}{5} \\ 1 \\ \frac{4}{5} \end{pmatrix}$$

10) again. orthog. refl. in ABC.

$$\bar{u}: 2x + z + 1 = 0.$$

$$\Rightarrow m = (2, 0, 1), \quad a_{\text{ntt}} = 1.$$

$$|a|^2 = 5.$$

$$a \otimes a = a \cdot a^T = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

$$\begin{aligned} I - 2 \cdot \frac{a \otimes a}{|a|^2} &= I - \frac{2}{5} \begin{pmatrix} 4 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 1 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} - \frac{2}{5} \begin{pmatrix} 4 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 1 \end{pmatrix} = \\ &= \frac{1}{5} \begin{pmatrix} 5-8 & 0-0 & 0-4 \\ 0 & 5-0 & 0-0 \\ 0-4 & 0-0 & 5-2 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -3 & 0 & -4 \\ 0 & 5 & 0 \\ -4 & 0 & 3 \end{pmatrix} \end{aligned}$$

$$\text{Ref} = \underbrace{\left(I_2 - 2 \cdot \frac{a \otimes a}{|a|^2} \right)}_{\text{lin. part.}} P - \underbrace{2 \cdot \frac{a_{\text{ntt}}}{|a|^2} \cdot a}_{\text{transl. part.}}$$

$$-2 \cdot \frac{1}{5} \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{4}{5} \\ 0 \\ -\frac{2}{5} \end{pmatrix} = \frac{1}{5} \begin{pmatrix} -4 \\ 0 \\ -2 \end{pmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{5} \begin{bmatrix} -3 & 0 & -4 \\ 0 & 5 & 0 \\ -4 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \frac{1}{5} \begin{bmatrix} -4 \\ 0 \\ -2 \end{bmatrix}$$

11) parallel projection. on ABC, $\parallel \vec{CD}$, $\bar{u}: 2x + z + 1 = 0$, $a = (2, 0, 1)$, $a_{\text{ntt}} = 1$.

$$P_{\text{ABC}, v}(P) = \left(I_m - \frac{v \cdot a^T}{v^T a} \right) P - \frac{a_{\text{ntt}}}{v^T a} v$$

$$v = \vec{CD} = D - C = (0, -1, 3)$$

$$\text{lin. part} = I_m - \frac{1}{3} \begin{pmatrix} 0 & 0 & 0 \\ -2 & 0 & -1 \\ 6 & 0 & 3 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 0 & 0 & 0 \\ -2 & 0 & -1 \\ 6 & 0 & 3 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 3 & 0 & 0 \\ 2 & 3 & 1 \\ -6 & 0 & 0 \end{pmatrix}$$

$$v \cdot a^T = \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} \begin{pmatrix} 2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ -2 & 0 & -1 \\ 6 & 0 & 3 \end{pmatrix}$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 3 & 0 & 0 \\ 2 & 3 & 1 \\ -6 & 0 & 0 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix}$$

$$v^T a = (0 -1 3) \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} = \cancel{3} \cancel{3} \cancel{3}$$

$$\text{transl. part} = \frac{1}{3} \begin{pmatrix} 0 \\ -1 \\ 3 \end{pmatrix} =$$

1) again. $(B, \vec{AB}, \vec{AC}, \vec{AD})$; proj. on ABC, $\parallel \vec{CD}$.

general point: (x', y', z')

$$\Rightarrow X = B + x' \cdot \vec{AB} + y' \cdot \vec{AC} + z' \cdot \vec{AD}, \quad B = (0, 0, 0).$$

$$ABC = \vec{AB} \times \vec{AC} \Rightarrow \text{no } \vec{AD} \Rightarrow z' = 0.$$

$$\vec{CD} = D - C = (0, -1, 3)$$

$$\vec{CD} = \alpha \cdot \vec{AB} + \beta \cdot \vec{AC} + \gamma \cdot \vec{AD}$$

$$(0, -1, 3) = \alpha(0, 2, 0) + \beta(1, 3, -2) + \gamma(1, 2, 1)$$

$$= (\beta + \gamma, 2\beta + 2\gamma, -2\beta + \gamma)$$

$$\Rightarrow 0 = \beta + \gamma \Rightarrow \gamma = -\beta.$$

$$-1 = 3\beta + 2\gamma \Rightarrow -1 = 3\beta - 2\beta = \beta \Rightarrow \boxed{\beta = -1}$$

$$3 = -2\beta + \gamma \Rightarrow 3 = -2(-1) - 1 = 1 \Rightarrow \boxed{\gamma = 1}$$

$$-1 = -1 + 2\alpha \Rightarrow \boxed{\alpha = 0}$$

$$\vec{CD} = 0 \cdot \vec{AB} - \vec{AC} + \vec{AD} = \vec{AD} - \vec{AC} = (1, 2, 1) - (1, 3, -2) = (0, -1, 3)$$

$$\Rightarrow \underline{(0, -1, 1)}$$

Line through P, $\parallel \vec{CD}$: $l: P + t \cdot (0, -1, 1)$.

$$l: (x', y', z') + t \cdot (0, -1, 1) =$$

$$\begin{cases} l: (x', y' - t, z' + t) \\ z' = 0 \end{cases}$$

$$\Rightarrow z' + t = 0 \Rightarrow t = -z'$$

$$\Rightarrow (x', y' + z', z' - z') =$$

$$\Rightarrow (x', y' + z', 0)$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + 2y - 1 = 0$$

$$= \text{Rot}_{0, 45^\circ} \circ \text{Ref}$$

$$0 + 2 \cdot 1 - 1 = 1$$

$$(1, 2) = (0, 1) - \frac{2}{5}$$

$$\text{Rot}_{0, 45^\circ} \circ (\text{Ref}_l(0))$$

$$+ by_0 + c = a \cdot 0 + b \cdot$$

$$2 \cdot \left(\frac{-1}{5}\right) (1, 2) = \frac{2}{5}$$

$$\begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

Quiz 5 - Redone

1. proj. are inv. FALSE

[B] homog. matrix in $\mathbb{R}^3$ is 4×4 . ✓

[C] refl. are inv. ✓

II. $[A]_K = (-1, -1, 1)$, $[B]_K = (-1, 1, 1)$, $[C]_K = (0, 2, -1)$, $[D]_K = (0, 1, 2)$
 7) orthog. proj. of D on plane containing ABC.

$$\begin{aligned} \vec{n}_{ABC} &= \vec{AB} \times \vec{AC} = (0, 2, 0) \times (1, 3, -2) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 2 & 0 \\ 1 & 3 & -2 \end{vmatrix} = \hat{i} \begin{vmatrix} 2 & 0 \\ 3 & -2 \end{vmatrix} + \hat{j} \begin{vmatrix} 0 & 0 \\ 1 & -2 \end{vmatrix} + \hat{k} \begin{vmatrix} 0 & 2 \\ 1 & 3 \end{vmatrix} \\ &= \hat{i}(-4) + 0 + (-2)\hat{k} = (-4, 0, -2) \\ \Rightarrow \vec{n}_{ABC} &= (2, 0, 1) \end{aligned}$$

$\Rightarrow$ plane through A with $\vec{n}_{ABC}$!

$$\pi: 2(x+1) + 0 \cdot (y+1) + 1 \cdot (z-1) = 0$$

$$\pi: 2x + z + 1 = 0.$$

$$\Rightarrow a_{n+1} = 1.$$

$$\vec{n}^T \cdot \vec{D} + 1 = (2 \ 0 \ 1) \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = 2 + 1 = 3$$

$$|\vec{n}|^2 = (\sqrt{2^2 + 0^2 + 1^2})^2 = (\sqrt{5})^2 = 5.$$

8) parallel refl. in ABC, $\parallel \vec{DA}$

$$\pi: 2x + z + 1 = 0$$

$$\vec{a} = (2, 0, 1), \quad a_{n+1} = 1.$$

$$\vec{DA} = A - D = (-1, -2, -1)$$

$$\text{Ref}_{\pi, \vec{v}}(D) = D - 2 \cdot \frac{\vec{a}^T D + a_{n+1}}{\vec{a}^T \vec{v}} \vec{v}$$

$$\vec{a}^T D = \dots$$

10) matrix form of orthog. refl. in plane π . (ABC)

$$\text{Ref}_{\pi}^{\perp}(P) = \left(I - 2 \cdot \frac{\vec{a} \otimes \vec{a}}{|\vec{a}|^2} \right) P - 2 \cdot \frac{a_{n+1}}{|\vec{a}|^2} \cdot \vec{a}$$

$$\text{lin. part} = \frac{1}{5} \begin{bmatrix} -3 & 0 & -4 \\ 0 & 5 & 0 \\ -4 & 0 & 3 \end{bmatrix}, \quad \text{transl. part} = -\frac{2}{5} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = +\frac{1}{5} \begin{bmatrix} -4 \\ 0 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \frac{1}{5} \begin{bmatrix} -3 & 0 & -4 \\ 0 & 5 & 0 \\ -4 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} + \frac{1}{5} \begin{bmatrix} -4 \\ 0 \\ -2 \end{bmatrix}.$$

11) Frame $(B, \vec{AB}, \vec{AC}, \vec{AD})$, proj. on plane ABC, $\parallel \vec{CD}$.

$$\vec{CD} = D - C = (0, -1, 3).$$

$$X = B + x' \vec{AB} + y' \vec{AC} + z' \vec{AD}, \quad B = (0, 0, 0) \quad A, B, C, D.$$

$$\vec{CD} = \vec{AD} - \vec{AC} \Rightarrow x' = 0, y' = -1, z' = 1.$$

$$N = [\vec{CD}]_B = (0, -1, 1) \leftarrow \text{to be used!}$$

$$[\vec{CD}]_{\text{init. frame}} = (0, -1, 3).$$

$$\vec{AB} \times \vec{AC} = \vec{0} \Rightarrow z' = 0.$$

$$0 \cdot x' + 0 \cdot y' + 1 \cdot z' = 0 \Rightarrow (0, 0, 1) = a; a_{n+1} = 0.$$

normal vector in basis B is not the same as in ex 10)!!!

(K1)

$$P_{\Pi}^{\parallel}(P) = \left(I - \frac{N \cdot N^T}{N^T N} \right) P - \frac{a_{n+1}}{N^T A} \cdot N$$

$$N \cdot A^T = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$N^T A = \begin{bmatrix} 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = 1$$

$$\text{lin. part} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{transl. part} = \frac{0}{1} \cdot \begin{bmatrix} 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \cdot 0.$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} + \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

(K2) line through P, $\parallel \vec{CD}$: $l: P + t \cdot (0, -1, 1)$

$$\begin{cases} l: (x', y', z') + t \cdot (0, -1, 1) \\ z' = 0 \end{cases}$$

$$\Rightarrow \begin{cases} l: (x' + y' - t + z' + t) = 0 \\ z' = 0 \end{cases}$$

$$\Rightarrow z' + t = 0 = t = -z'.$$

$$l: (x', y', z') - z' \cdot (0, -1, 1) = (x', y' + z', z' - z') = (x', y' + z', 0)$$

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} \mapsto \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$